SEAGAS - Team Update: What Is Done, What Is Left, What To Do Next

From Carl (Soh Way Miin), Backend Developer, to Group 10 | 18 October 2026
COS40005 Computing Technology Project A | POC004 | Sprint 2 report due 24 October

In short: advisors can now open a full record for each student; students have a Progress page and can attach photos or PDFs as proof to their projects and certifications; admins can manage the learning resources the recommendation engine suggests. After you pull, run **start_all.bat** once (section 3). No new packages, and you do not have to sign in again.

Note: these changes were pushed straight to dev, without a pull request. Please look through the parts that touch your work and tell Carl about anything that looks wrong.

1. What Is Done (18 October)

1.1 New features (commit f157635, now on dev)

Area	What changed
Advisor: student detail	Click a student in the Students tab to open their record: profile, latest assessment with the 7 dimension scores, Strong / Developing / Gap skills (reason on hover), readiness over time, skills, projects and certifications with their proof files, and the recommended activities they started or finished.
Advisor: student list	Search by name, email, programme or role; filter by On Track / Developing / At Risk / No Assessment; shows 100 rows at a time ("Show more"). A real score of 0% no longer shows as "No Assessment".
Advisor: Profiles Complete	Was counting every profile (5,000). It now counts complete profiles only (3,823 on the synthetic data). The backend never set `is_complete` for real students; it now does whenever a student saves the profile or adds / removes a skill. Complete = programme, university, year of study and target role filled in, and at least one skill.
Student: Progress page	New menu item **Progress**: current readiness, change since the first assessment, completed activities, a readiness-over-time chart per role, and the lists of activities in progress and completed. For each activity in progress: **Mark Complete**, **View in Recommendations** and **Give up** (it goes back to "not started"; completed activities cannot be removed).
Student: projects and certifications	Each one can now be edited and deleted. Errors are shown on the page instead of being ignored.
Student: proof files	Attach JPG, PNG or PDF files (max 5 MB each, 5 per project / certification), either in the Add / Edit form or later under each entry. The file type is checked from the file's content, so a renamed .html or .svg is refused. Students see only their own files; advisors and admins can open any, and each time is written to the Audit Log ("Opened evidence file").
Admin: Learning Resources	New page listing all 299 resources with how many students started / finished each. Admins can add resources and switch any on or off (inactive ones are no longer recommended). Same rule as skills and roles: seed resources are only switched on/off; resources added by an admin are editable.
Look and feel	Dropdowns and badges show capitalised text (Personal, Beginner, AI / Digital, Course ...). The "Skill added" marks were removed; completed activities just show Completed.

Tests	`tests/test_backend_api.py` : 43 API tests (8 new) covering all of the above, including file type checks, the 5-file limit, ownership and the audit entry. All pass.

1.2 Database changes

Change	Why
New table `student_evidence_files`	Proof files. Each belongs to one project or one certification and is deleted with it. The file bytes are in the `data` column: never select it in list queries.
`recommendations_library.source`	`seed` or `admin` , like `job_roles.source` .
`student_profiles.is_complete` recomputed	For real (non-synthetic) students. Synthetic students keep their generated value.

All three are in both `database/schema.sql` (new databases) and `database/migrate_existing_db.sql` (existing ones). **start_backend.bat** runs the migration on every start, so you do not run anything by hand.

2. Project Status: the 13 Deliverables

Rows marked "no change" are as reported on 16 October; I did not re-check them today.

#	Deliverable	Status	Notes
1	Working web platform	**Mostly done**	Three portals; new Progress, Student Detail and Learning Resources pages.
2	Student profile	**Partial**	New: edit / delete, proof files. Missing: work experience details, evidence choice when adding a skill, student number (decision needed, 4.2).
3	JD NLP module	**Done**	No change.
4	AI skill matching	**Done**	No change.
5	Readiness score	**Done**	No change.
6	Skill gap analysis	**Done**	No change.
7	Recommendation engine	**Done**	Admins can now manage the resources.
8	Student dashboard	**Done**	No change.
9	Advisor dashboard	**Mostly done**	New: student detail, search / filter, correct Profiles Complete. Missing: advisor notes / review of AI results, report export, readiness histogram, cohort progress view.
10	Development tracking	**Done**	New Progress page; complete or give up from there. Time taken is stored but not shown (4.2).
11	AI evaluation	**Partial**	No change: no gold labels yet, so no precision / recall / F1.
12	Usability (SUS)	**Planned**	No change.
13	Docs, report, demo	**Partial**	Sprint 2 report due 24 October.

Not started (from the Team Plan): **Career Paths** page and API (FR-15, Low priority), **System Settings** for admins (FR-16), optional CV upload.

3. What To Do Next

3.1 After you pull (every member)

1. `git checkout dev` then `git pull`.
2. Double-click **start_all.bat**. The backend window updates your database by itself (new table and column, recomputed profiles). No new packages are needed: uploads use `python-multipart`, already in `backend/requirements.txt`.
3. Refresh the browser (Ctrl+F5) and click through the new pages:
   - Student: **Progress**; My Profile, Projects / Certifications (edit, delete, attach a proof file).
   - Advisor: Students tab, then click a student.
   - Admin: **Learning Resources**; the Audit Log after an advisor opens a proof file.
4. Use only made-up certificates and photos when testing. Real ones contain names and ID numbers, and the Team Plan (NFR-05) says the POC uses no real student data.

3.2 Before your next push

1. **Pull dev first** and merge it into your branch, then run the site once. The files below changed a lot, so expect merge conflicts if you edited them:

File	Most likely affects
`frontend/src/pages/Profile.jsx`	Ren Hang, Shanice
`frontend/src/pages/AdvisorDashboard.jsx`, `Recommendations.jsx`	Ren Hang, Aaron
`frontend/src/api/client.js`, the three `portals/*Portal.jsx`	Anyone adding pages or API calls
`backend/routers/students.py`, `advisor.py`, `progress.py`, `admin.py`	Anyone changing the backend
`database/schema.sql`, `migrate_existing_db.sql`	Yong Hin, Carl

2. Run the tests: `python -m pip install -r tests\requirements.txt` then `python -m pytest tests/test_backend_api.py -q` (43 should pass; they create and delete their own database).
3. Run `npx vite build` in `frontend` to catch broken imports.
4. Changing the database? Put the change in **both** `schema.sql` and `migrate_existing_db.sql` (with `IF NOT EXISTS`), otherwise teammates with an existing database will not get it.
5. New dropdowns or badges: show the text through `label()` from `frontend/src/utils/labels.js` (CSS `text-transform` does not reach an opened dropdown list).
6. Commit messages in English, and push through a pull request into dev.

3.3 If something goes wrong

Problem	Fix
Backend error about `student_evidence_files` or `source`	The migration did not run. Start with start_backend.bat, or run `psql -d seagas_db -f database\migrate_existing_db.sql`.
"Only JPG, PNG or PDF files can be attached"	The file's content is not one of these types, even if the name ends with .png. Save it again as JPG, PNG or PDF.
"View" opens nothing	The browser blocked the new tab. Allow pop-ups for localhost:5173.
Advisor still sees 5,000 Profiles Complete	Restart the backend so the new code runs; refresh the page.

4. What Is Still Left

4.1 Tasks

Owners follow the Team Plan roles and the 16 October update; change them at the Thursday meeting if needed.

Owner	Task	When
Everyone	Pull, run start_all.bat once, check your own pages (3.1). Sprint 2 report.	Now / 24 Oct
Yong Hin	Gold labels and precision / recall / F1; NLP alias fixes; 30 JDs per role (from the 16 October list).	Sprint 3-4
Ren Hang, Shanice	Evidence choice when adding a skill; "decision support" notice; "At Risk" to "Needs support"; empty catch blocks left in Profile.jsx (3 places).	Sprint 3
Aaron	Advisor notes / review of AI results (human oversight); Career Paths API and page (FR-15).	Sprint 3
Carl	Advisor report export with Ren Hang; readiness histogram and cohort progress view on the advisor overview; work-experience details; integration and bug fixes.	Sprint 3
Not assigned	Admin System Settings (FR-16); optional CV upload; "Give up" button on the Recommendations page too.	Sprint 3-4
Whole team	SUS usability test; final report and demo.	Sprint 4

4.2 Decisions needed at the next meeting

Question	Options and notes
Student number	The system has no student ID, only email and an internal user ID. Proposal: add `users.student_number` (unique), entered at registration, editable only by admins, shown and searchable for advisors and admins; fake numbers for the 5,000 synthetic students. Needs the INTI number format, and touches the Register page (Ren Hang) and the synthetic data generator (Yong Hin). It is personal data, so the report should say why it is needed (APP 3).
Editing the 6 seed job roles in the Admin page	Today they are read-only because `seed_job_role_skills.sql` rewrites their skills on every start, so admin edits would be lost. Making the database the source of truth needs Yong Hin's agreement: later seed changes would no longer reach existing databases.
Show how long activities took	Start and complete times are already stored. Options: (1) show "took N days" on the Progress and Student Detail pages; (2) average days per resource for admins; (3) let students enter hours spent. Note: the stored time is between the two clicks, not real study time.
"Give up": delete or keep	Giving up now deletes the activity, so advisors cannot see it. The alternative is a `dropped` status that advisors can see.